CSC111 Project Proposal: A list of recommended games on steam

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Problem Description and Research Question

As game lovers, we often find it difficult to quickly discover a game that truly fits our preferences. This is especially true for students, who have limited time to browse through thousands of Steam pages, compare information, and read lengthy reviews. While Steam offers a vast catalogue and a tag system for searching, the process of looking for a "good" game can still feel overwhelming, even when we already have a specific genre of game in mind. Our tool will help users discover games that match their preferred genres and budget without a big time investment. For this project, we want to build a recommendation system that's fast, simple, and budget-friendly for students. Instead of only relying on basic filtering, we also want to recommend games that are actually similar to what the user already enjoys, using features like tags and platforms to make the results feel more relevant.

Our central question is: **How can we design an interactive system that recommends Steam games to users based on their preferences?** The goal of this project is to create an interface that prompts users for their tastes (e.g., preferred genres, price range, and how strict they want the reviews to be) and then generates a personalized list of recommended games, complete with average ratings and other key statistics. To make this recommendation process more transparent, we will model relationships between games using a graph built with `networkx`. In this graph, each game is a node, and we connect two games with an edge if they are similar based on tag overlap, platform overlap, and review information. This lets us recommend games by starting from games the user already likes and then exploring nearby connected games, instead of just showing the most popular games in a genre. We also want the recommendations to be explainable, so users can click a suggestion and see the "path" or connection (shared tags/features) that led to the list of recommended games on Steam.

Computational Plan

Our project will model the Steam ecosystem using a **graph-based data structure**. We will create two primary classes: `Game` and `User`.

The `Game` class will represent individual games, with attributes like `app_id`, `name`, `price`, `genres`, `platforms`, `tags`, and review statistics. The core of our project is a **game similarity graph**, where each `Game` object is a vertex. An edge will be added between two games if their calculated similarity score exceeds a threshold. This score will be computed using a weighted combination of Jaccard similarity on their genre and tag sets, and a binary score for platform overlap (e.g., $0.8 \times \text{tag\_similarity} + 0.2 \times \text{platform\_match}$). This graph allows us to move beyond simple filtering and find games that are conceptually related.

The `User` class will represent a user and will be connected to the `Game` objects if they have reviewed or commented on one certain game, forming a **user-game graph**. This structure will enable more advanced recommendation features in the future, such as collaborative filtering (e.g., "users who liked this game also liked..."). User could also view comments of other users to the game.

To meet the requirement for a tree-based structure, we will also implement a **genre tree**. This tree will organize games hierarchically (e.g., "Action" as a root with children "Shooter" and "Fighting," which further branch into specific games). This will allow users to navigate and explore the game catalogue in a more structured way, or allow our algorithm to recommend from broader or narrower categories based on their input.

We will use the **`networkx, tkinter,matplotlib` library**. `Networkx`, a new library for us, to help construct, analyze, and visualize our `Game-User` graph. We plan on showing how our algorithm works and show the path that was taken to determine the recommended list of Steam games based on user input. Besides that, `matplotlib` would be the external library which helps us draw the graph/subgraph or show charts of ratings/genres.

Our program will be interactive, built with `tkinter`. The interface will be a window opened where we will ask users a series of questions to gather their preferences, such as:

- Preferred genres (selected from the genre tree)
- Budget (Maximum price, Minimum Price)
- Minimum acceptable positive review ratio

then it will display the recommended games as a list, there will also be a button for the user to click (using the Button attribute within tkinter) that can direct them towards the graph visualization.

The program will then perform a multi-step search:

1. Filter games by the user's genre and price preferences.
2. Continue to filter these by their positive review ratio.
3. Use the game similarity graph to find games that are highly connected to the top results, expanding the recommendation set.
4. Rank the final list of games using a metric that combines popularity (e.g., number of reviews) and similarity score.
5. Display the top 10 recommendations to the user in the `tkinter` interface, showing each game's name, price, and average rating. The user could then click on a game to see its similarity graph neighborhood, visualized with `networkx` and `matplotlib`.

Dataset: We will use a real-world dataset of Steam games and user reviews. Specifically, the "Steam Games Review" and "Steam Game Dataset" available on Kaggle and Github. This dataset contains the attributes (genre, tags, price, reviews) necessary to build our graphs and perform our computations.

References by Category

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